Indian Institute of Technology Bhubaneswar

Department of Computer Science and Engineering

School of Electrical and Computer Sciences

Deep Learning (CS6L038)

Date: April 9, 2026

Class Test (Spring 2025-26)

Time: 45 Mins ***Use standard notations as discussed in class** Maximum Marks: 10

NAME: **Roll No.:**

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1. Given the mini-batch $\mathbf{u} = \begin{bmatrix} 7 & 13 \\ 9 & 15 \\ 8 & 11 \end{bmatrix}$, compute the batch-normalized output for each feature (column-wise). Assume $\gamma = [1.5, 0.5], \quad \beta = [2, -1], \quad \epsilon = 10^{-7}$. (3)

2. An input image of size $64 \times 48 \times 3$ is passed through the following CNN architecture: (3)
   - Convolutional Layer 1: 8 filters of size 3×3, stride 1, padding 1
   - Max Pooling Layer 1: 2×2 with stride 2
   - Convolutional Layer 2: 16 filters of size 5×5, stride 1, padding 2
   - Max Pooling Layer 2: 2×2 with stride 2

   (a) Compute the spatial size of the feature maps after each layer.
   (b) Compute the total number of learnable parameters in the network.

3. Consider a single-hidden-layer MLP with the following parameters: (2)

   Input $\mathbf{x} = [1,\ 2]^\top, \quad W_1 = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, \quad b_1 = [0,\ 1]^\top, \quad W_2 = [1,\ -2]^\top, \quad b_2 = 1$

   Assume the forward pass is given by $z = W^\top \mathbf{x} + b$. The hidden layer uses *tanh* activation, and the output layer uses *sigmoid* activation. Compute the final output of the network.

4. Why is Adam optimizer often preferred in deep learning? (1)

5. What are the key inductive biases in Convolutional Neural Networks (CNNs)? (1)

Write your answer in the space below.

1 mini-batch $u = \begin{bmatrix} 7 & 13 \\ 9 & 15 \\ 8 & 11 \end{bmatrix}$ → feature 1 → feature 2

Step-1: Compute the mean (feature-wise)

$\mu_1 = \frac{7+9+8}{3} = 8$ $\quad \mu_2 = \frac{13+15+11}{3} = 13$

$$\boxed{\mu = \frac{\sum x_i}{n}}$$

Step-2: Compute the variance (feature-wise)

$$\sigma_1^2 = \frac{(7-8)^2 + (9-8)^2 + (8-8)^2}{3} = 2/3$$

$$\sigma_2^2 = \frac{(13-13)^2 + (15-13)^2 + (11-13)^2}{3} = 8/3$$

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

In batch Normalization, always n is used to calculate the variance

Step-3: Normalize

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

(ϵ will be used when σ is zero).

$\sigma_1 = \sqrt{2/3} \approx 0.816$

$\sigma_2 = \sqrt{\frac{8}{3}} \approx 1.633$

Normalized values:

feature-1: $\begin{bmatrix} \frac{7-8}{0.816} \\ \frac{9-8}{0.816} \\ \frac{8-8}{0.816} \end{bmatrix} = \begin{bmatrix} -1.225 \\ 1.225 \\ 0 \end{bmatrix}$

feature 2: $\begin{bmatrix} \frac{13-13}{1.633} \\ \frac{15-13}{1.633} \\ \frac{11-13}{1.633} \end{bmatrix} = \begin{bmatrix} 0 \\ 1.225 \\ -1.225 \end{bmatrix}$

Step-4: Scale and shift

given $\gamma = [1.5, 0.5]$ $\quad \beta = [2, -1]$

Apply $u = \gamma \hat{x} + \beta$

$$\therefore u = [1.5 \;\; 0.5] \odot \begin{bmatrix} -1.225 & 0 \\ 1.225 & 1.225 \\ 0 & -1.225 \end{bmatrix} + [2 \;\; -1]$$

point-wise multiplication along each row

$$= \begin{bmatrix} 0.1625 & -1 \\ 3.8375 & -0.3875 \\ 2 & -1.6125 \end{bmatrix}$$

(final Answer)

(2) (a) input size : $64 \times 48 \times 3 = W \times H \times 3$

O/p size after $K \times K$ convolution with stride s and padding p is given by

$$\left[\left\lfloor \frac{W-K+2p}{s} \right\rfloor + 1 , \left\lfloor \frac{H-K+2p}{s} \right\rfloor + 1 \right]$$

(i) Layer-1 : $\left(\left\lfloor \frac{64-3+2}{1} \right\rfloor + 1 , \left\lfloor \frac{48-3+2}{1} \right\rfloor + 1 \right)$ $= 64 \times 48$

(3x3 conv, stride 1, padding 1)

output : $\underline{64 \times 48 \times 8}$ → 8 filters

(ii) Layer-2 : max-pool (2x2, stride 2)

output : $\underline{32 \times 24 \times 8}$

(iii) Layer-3 : 5x5 conv, stride 1, padding 2

output : $\underline{32 \times 24 \times 16}$ → 16 filters.

(iv) Layer 4 : max-pool (2x2), stride 2.

output : $\underline{16 \times 12 \times 16}$

(b) Learnable parameters: (3)

convolution size

Layer 1: $8 \times (3 \times 3 \times 3 + 1) = 8 \times 28 = 224$

bias

no. of kernels / filters

no. of channels in the i/p

Layer 2: 0 (pooling doesn't involve any parameter)

Layer 3: $16 \times (5 \times 5 \times 8 + 1) = 16 \times 201 = 3216$

Layer 4: 0 (same as layer 2, as it is a pooling layer)

Total parameters: $224 + 3216 = \underline{3440}$

(3) $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

<u>Hidden layer:</u> $z_1 = W_1^T x + b_1$

$W_1 = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\quad = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$

After applying activation: $a_1 = \tanh(z_1)$

$$= \begin{bmatrix} \tanh(5) \\ \tanh(0) \end{bmatrix} \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

<u>Output layer:</u>

$$z_2 = W_2^T a_1 + b_2$$

$W_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad b_2 = 1$

$$\therefore z_2 = \begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 = 2$$

After applying activation: $a_2 = \sigma(z_2)$

$$= \sigma(2) \approx 0.88$$

Final output: $\underline{0.88}$ (approx.)

4. Adam optimizer is often preferred in deep learning, because
   (i) it combines momentum and adaptive learning rate
   (ii) Faster convergence in practice

5. Inductive biases in CNN :
   (i) Locality : Nearby pixels are related
   (ii) Invariance to translation : Detects features regardless of position
   (iii) Hierarchical compositionality : Simple features are combined together to form complex features.

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